33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F22-SR-001
415V CARBON TREATMENT & ELUTION SWITCHROOM

F22-TR-030
TRANSFORMER ROOM

F22-TR-030
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
4000/ 1A
2.5VA
5P20

F22-SW-030
CARBON TREATMENT & ELUTION 415V MOTOR CONTROL CENTRE No.1

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
Cl 1
4000/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
ACB
AFD
4000A
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6
CONTINUED ON SHT 2
2000-F22-SLD-EL-10202

415V 3NE~ 50Hz 4000A 65kA for 1sec

1F-01	2F-11	3F-11	1F-13	4R-01	3R-11	INC	2R-11	4R-13	4F-05	2F-08	2R-05	3R-01											
630A	630A	630A	630A	630A	630A		630A	630A	250A	63A	100A	160A											
2A 32A																							
250A	250A	250A	250A	250A	250A		250A	250A	90A	70A	100/1A	100/1A											
	EOL	EOL			EOL		EOL		EOL														
3PX			3PX	3PX				3PX															
168kW	168kW	168kW	168kW	168kW	168kW	168kW	168kW	168kW	75kW	22.5kW	30kW	45kW											
F22-KN-411A	F22-KN-411B	F22-KN-411C	F22-KN-411D	F22-KN-411E	F22-KN-411F	F22-KN-411G	F22-KN-411H	F22-PP-460	F22-HT-411	F22-PP-490	F22-PW-441												
CIL4	CIL4	CIL4	CIL4	CIL4	CIL4	CIL4	CIL4	CIL4	CIL4	CIL4	CIL4												
CARBON REGENERATION KILN HEATERS ZONE 1 BANK 1	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 2	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 3	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 4	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 5	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 6	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 7	CARBON REGENERATION KILN HEATERS ZONE 1 BANK 8	ELUTION HEATER THERMAL OIL PUMP	ELUTION HEATER	BARREN CARBON PUMP	ELUTION PUMP 1												
2000-F22-SCM-EL-10207	2000-F22-SCM-EL-10208	2000-F22-SCM-EL-10209	2000-F22-SCM-EL-10210	2000-F22-SCM-EL-10211	2000-F22-SCM-EL-10212	2000-F22-SCM-EL-10213	2000-F22-SCM-EL-10214	2000-F22-SCM-EL-10223	2000-F22-SCM-EL-10225	2000-F22-SCM-EL-10224	2000-F22-SCM-EL-10227												

F22SR001-LCS
NOTE 12

Approved	☒
Approved as Noted	☐
Not Approved	☐
For Information Only	☐

Signed ■ Date 06/01/2026

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F22-SW-030 INCOMER TIER." CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF THE SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F22-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET
- 3PX — THYRO POWER CONTROLLER THYRO-PX-3PX-500-350-H-F WITH TOUCH DISPLAY OPERATED IN TAKT MODE (ZERO CROSS FIRING)

DRG No.	REFERENCES
2000-F22-SCM-EL-10204	F22-SW-030 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F22-SCM-EL-10203	F22-SW-030 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F22-SCM-EL-10201	F22-SW-030 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F22-DRG-EL-10101	F22-SR-001 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F22-BLK-IC-10001	F22-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F22-SLD-EL-10202	F22-SW-030 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	24NOV25	FAT REDLINES INCORPORATED					
0	05NOV24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: CIL4 CARBON TREATMENT & ELUTION
F22-SW-030 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F22-SLD-EL-10201	1

O:\CAD PROJECTS\23S NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F22 - CIL4 CARBON TREAMENT & ELUTION\2000-F22-SLD-EL-10201.DWG 18DEC25 12:14PM HOTSEAT04